

TUGATAI FINAL AI SAAS MASTER PRODUCT + SYSTEM ARCHITECTURE

1. Product Vision

Tugatai AI SaaS is not just a chatbot platform.

It is a **multi-channel AI business operating system** designed to help businesses attract leads, answer customers, generate quotes, create bookings, automate follow-ups, manage CRM, and scale operations.

It should be usable first for **Tugatai Cargo**, but built in a way that can later support other industries such as:

- restaurants
- laundry businesses
- law firms
- schools
- clinics
- service companies

The product should combine the strengths of:

- **Moonline** for landing page, voice AI positioning, and demo-first marketing
 - **WIRAL** for SaaS pricing structure, feature comparison, and conversion flow
 - **Tugatai** for real business workflow: quote → booking → scheduling → tracking → CRM
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2. Core Product Positioning

Competitor positioning

- **WIRAL** = chat + automation
- **Moonline** = voice AI
- **Tugatai AI SaaS** = AI Chat + Voice + Quote + Booking + CRM + Operations

Main promise

The system should move businesses from:

inquiry → engagement → quote → booking → conversion → retention

Internal business flow

Inside the platform, the main operational flow should be:

Chat / Voice → AI Understanding → Quote → Booking → Scheduling → Tracking → CRM

3. Landing + Conversion Layer

This is the public-facing SaaS website and onboarding entry.

Hero section

The top section should be **Moonline-style**:

- strong product headline
- short benefit-driven subtext
- demo-first CTA approach

Suggested hero messaging

- **Headline:** AI Voice & Chat Agent for Businesses
- **Subtext:** Handles chats, calls, bookings, follow-ups, and support 24/7

CTAs

- Start Free Trial
- Schedule Demo
- Try AI (Chat / Voice)

Goal

When visitors land on the page, they should quickly understand the value and either:

- test the AI
- request a demo
- start a free trial

Marketing flexibility

This top section should support different industries:

- cargo
- restaurants

- law firms
 - laundry
 - schools
 - others
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4. Pricing / Subscription Model

The system should use branded pricing tiers.

Plan names

- **Half Moon**
- **Full Moon**
- **Moon Enterprise**

Plan logic

These plans should control:

- users
- channels
- chats/messages
- AI depth
- CRM depth
- booking features
- pricing engine access
- automation depth
- campaigns
- voice AI
- integrations
- multi-branch support

Half Moon

For small businesses / entry plan:

- 2 users
- 2 channels
- up to 1,000 chats/month
- shared inbox
- basic AI instant replies
- basic lead capture

- basic booking form
- limited analytics
- no advanced automation
- no campaigns
- no voice AI

Full Moon

Main commercial plan:

- 5–10 users
- multi-channel
- up to 5,000 chats/month
- everything in Half Moon, plus:
- advanced AI assistant
- CRM pipeline
- full booking workflow
- pricing engine
- quote generation
- workflow automation
- chat summaries
- WhatsApp campaigns (basic)
- analytics dashboard
- AI assist for agents

Moon Enterprise

Advanced / high-volume plan:

- unlimited or high-volume chats
- unlimited users/channels
- everything in Full Moon, plus:
- multi-branch / multi-country
- advanced workflow automation
- advanced campaigns
- API integrations
- advanced analytics
- dedicated onboarding
- priority support
- voice AI included or as premium add-on

Important separation

Subscription plans must be separate from business quote logic:

- plans control feature access and limits
 - pricing engine controls business/service pricing
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5. Customer Channels

The platform should support omnichannel communication.

Current channels

- WhatsApp Cloud API
- Instagram Messaging API
- Facebook Messenger API
- Website Live Chat
- Voice channel

Future channels

- Telegram
 - LinkedIn
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6. Channel Connector Layer

This layer connects all channels into one normalized system.

Main functions

- webhook receiver per channel
- inbound message capture
- message normalization
- outbound dispatcher
- attachment/media handling
- voice adapter for speech input/output

For voice:

- speech-to-text
 - text-to-speech
 - call event handling
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7. Unified Inbox Layer

This is the operational heart of the product.

Core features

- shared omnichannel inbox
- AI + human reply interface
- AI assist suggestions
- conversation history
- notes
- tags
- agent assignment
- AI/human switch control
- lead status updates

Status system

Recommended stages:

- cold
- warm
- hot
- quoted
- booked
- completed

Inline actions inside chat

The inbox should allow direct actions such as:

- generate quote
- create booking
- schedule pickup
- assign agent
- update lead stage
- send follow-up
- mark completed

Right-side structured panel

Should display:

- customer/contact info

- item/service details
 - pickup area
 - destination
 - weight
 - date/time
 - current status
 - recent summary
 - booking info
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8. Orchestration Layer

This is the decision/control layer for the product.

Key functions

- conversation router
- tenant resolver
- intent router
- AI trigger rules
- escalation rules
- workflow engine
- event/action engine

Workflow engine examples

- quote workflow
- booking workflow
- scheduling workflow
- follow-up workflow
- payment reminder workflow
- delivery notification workflow

This is what moves the system beyond a simple chatbot.

9. CRM / Lead Pipeline

Every new conversation should be usable as a lead.

CRM capabilities

- lead auto-creation
- contact profiles
- lead stage management
- reminders
- notes
- activity timeline
- conversion tracking

Sales stages

The CRM should support:

cold → warm → hot → quoted → booked → completed

This should update manually and automatically based on user activity and AI actions.

10. AI Intelligence Layer

This is the AI brain of the product.

Responsibilities

- intent detection
- entity extraction
- knowledge retrieval
- response generation
- conversation memory
- booking flow guidance
- quote assistance
- multilingual support

Extracted entities

Examples:

- destination
- pickup location
- weight
- item type
- date/time
- customer preferences

Voice AI intelligence

For voice mode, the system should also support:

- speech understanding
 - real-time response generation
 - call flow guidance
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11. Business Logic Layer

This is the biggest Tugatai advantage over generic AI tools.

Pricing engine

Supports:

- per kg pricing
- fixed item pricing
- TV pricing
- carton/drum/bicycle pricing
- CBM conversion
- customs fee
- promotions and discounts

Booking engine

Supports:

- pickup scheduling
- slot validation
- availability checks
- booking confirmation

Additional logic

- validation engine
 - rules engine
 - product/service logic
 - reference ID generation
 - shipment/order status handling
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12. Data Layer

PostgreSQL

Tables/entities should include:

- tenants
- users
- agents
- conversations
- messages
- leads
- bookings
- pricing rules
- products/services
- subscription plans
- usage logs
- audit logs

Vector database / knowledge base

For:

- FAQs
- policies
- service docs
- company-specific data

Redis / queue layer

For:

- async jobs
- retries
- rate limiting
- scheduled follow-ups
- automation jobs

13. Admin + Client Applications

Main apps

- Super Admin Dashboard
- Client Business Dashboard

- Unified Inbox UI
- CRM Dashboard
- Analytics Dashboard
- Knowledge Base Manager
- Pricing Manager
- Workflow / Automation Builder
- Campaign Manager
- Voice AI Settings
- Billing & Subscription Manager

Campaign manager

Should include:

- WhatsApp campaigns
- broadcast messages
- segmentation
- automation triggers

Analytics

Should show:

- chats
 - leads
 - bookings
 - conversions
 - revenue insights
 - team performance
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14. Mobile App

The mobile app should focus on operations.

Functions

- agent login
- inbox access
- reply to chats
- AI assist replies
- lead updates
- notifications

- basic booking / CRM actions
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15. Development Strategy

The full architecture should exist in planning, but development should happen in phases.

Phase 1

- landing page
- pricing page
- signup/onboarding
- omnichannel inbox
- AI replies
- CRM basics
- booking flow
- pricing engine
- analytics basics

Phase 2

- workflow automation
- campaigns
- advanced CRM
- multi-tenant improvements
- advanced reporting

Phase 3

- voice AI
- advanced integrations
- multi-branch / enterprise
- API layer
- industry templates

TUGATAI AI SAAS

— 1. LANDING / CONVERSION LAYER

- Hero section
 - Headline: AI Voice & Chat Agent for Businesses
 - Subtext: chats, calls, bookings, follow-ups 24/7
 - CTA buttons
 - Start Free Trial
 - Schedule Demo
 - Try AI (Chat / Voice)
- Demo-first experience
 - Live chat demo
 - Voice AI demo
 - Industry templates
- Pricing section
 - Half Moon
 - Full Moon
 - Moon Enterprise
- Conversion tools
 - Lead capture form
 - Demo booking
 - Signup / onboarding

— 2. SUBSCRIPTION + BILLING LAYER

- Plan management
 - Half Moon
 - Full Moon
 - Moon Enterprise
- Feature gating
 - users limit
 - channels limit
 - chats/messages limit
 - AI depth
 - CRM access
 - booking features
 - pricing engine access
 - automation depth
 - campaigns
 - voice AI
 - API access
 - multi-branch support
- Billing
 - free trial
 - subscription activation
 - upgrade / downgrade

- └ billing history
- └ payment status

└ 3. CUSTOMER CHANNELS

- └ WhatsApp Cloud API
- └ Instagram Messaging API
- └ Facebook Messenger API
- └ Website Live Chat
- └ Voice Channel
- └ Telegram (Phase 2)
- └ LinkedIn (Future)

└ 4. CHANNEL CONNECTOR LAYER

- └ Webhook receiver
- └ Message normalizer
- └ Outbound dispatcher
- └ Media/attachment handler
- └ Voice adapter
 - └ speech-to-text
 - └ text-to-speech
 - └ call events

└ 5. UNIFIED INBOX LAYER

- └ Shared inbox
- └ AI + Human reply interface
- └ AI Assist suggestions
- └ Agent assignment
- └ Conversation history
- └ Notes / tags
- └ Status management
 - └ cold
 - └ warm
 - └ hot
 - └ quoted
 - └ booked
 - └ completed
- └ AI / Human switch control
- └ Inline actions
 - └ generate quote
 - └ create booking
 - └ schedule pickup
 - └ update lead stage
 - └ send follow-up
 - └ mark completed

— 6. ORCHESTRATION LAYER

- Conversation router
- Tenant resolver
- Intent router
- AI trigger rules
- Human escalation rules
- Workflow engine
 - quote workflow
 - booking workflow
 - scheduling workflow
 - follow-up workflow
 - payment reminder workflow
 - delivery notification workflow
- Event/action engine

— 7. CRM / LEAD PIPELINE

- Lead auto-creation
- Lead stages
 - cold
 - warm
 - hot
 - quoted
 - booked
 - completed
- Follow-up reminders
- Conversion tracking
- Contact profiles
- Activity timeline
- Notes / internal updates

— 8. AI INTELLIGENCE LAYER

- Intent detection
- Entity extraction
 - destination
 - pickup location
 - weight
 - item type
 - date/time
- Knowledge retrieval
- Response drafting/generation
- Conversation memory
- Booking flow guidance
- Quote assistance

- └ Multi-language support
- └ Voice AI intelligence
 - └ speech understanding
 - └ real-time call response
 - └ call flow guidance

└ 9. BUSINESS LOGIC LAYER

- └ Pricing engine
 - └ per kg rates
 - └ fixed item pricing
 - └ CBM conversion
 - └ customs fee
 - └ promotions / discounts
- └ Booking engine
 - └ pickup scheduling
 - └ slot validation
 - └ availability checks
 - └ booking confirmation
- └ Validation engine
- └ Rules engine
- └ Product / service logic
- └ Reference ID generator
- └ Shipment / order state handling

└ 10. DATA LAYER

- └ PostgreSQL
 - └ tenants
 - └ users
 - └ conversations
 - └ messages
 - └ leads
 - └ bookings
 - └ pricing rules
 - └ products/services
 - └ subscription plans
 - └ usage logs
 - └ audit logs
- └ Vector DB / KB storage
 - └ FAQs
 - └ policies
 - └ service docs
 - └ business-specific knowledge
- └ Redis / Queue
 - └ async jobs

- └─ retries
- └─ rate limiting
- └─ scheduled follow-ups
- └─ automation jobs

└─ 11. ADMIN + CLIENT APPLICATIONS

- └─ Super Admin Dashboard
- └─ Client Business Dashboard
- └─ Unified Inbox UI
- └─ CRM Dashboard
- └─ Analytics Dashboard
 - └─ chats
 - └─ conversions
 - └─ bookings
 - └─ revenue insights
- └─ Knowledge Base Manager
- └─ Pricing Manager
- └─ Workflow / Automation Builder
- └─ Campaign Manager
 - └─ WhatsApp campaigns
 - └─ broadcast messages
 - └─ segmentation
- └─ Voice AI Settings
- └─ Billing & Subscription Manager

└─ 12. MOBILE APP

- └─ Agent login
- └─ Inbox access
- └─ Reply to chats
- └─ AI Assist replies
- └─ Lead updates
- └─ Notifications
- └─ Basic booking / CRM actions